GeoNews — Week 46/2026

Weekly digest on Quaternary palaeoclimate research, East African Rift dynamics, palaeohydrology and time-series analysis · 9 November 2026

> **What is GeoNews?** An **automatically generated weekly digest**. A research workflow scans publications from the past one to two weeks as well as funding, policy and community sources, scores the hits against a fixed set of criteria and assembles this issue from them.
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> The thematic profile is **aligned with the research interests of Markus L. Fischer** (Palaeoclimate Dynamics group, University of Potsdam): Quaternary palaeoclimate and palaeoenvironmental reconstruction, the East African Rift, the African Humid Period and human–environment coupling, lake-level and water-balance modelling, palaeofloods and fluvial geomorphology, ICDP drill cores, and linear and nonlinear time-series analysis. Researchers in these fields will probably find something useful here — it is explicitly not a complete overview of Quaternary science.
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> Selection, scoring and summaries are generated automatically. Entries may contain errors; when in doubt, the linked original source prevails.

Each paper receives a score of 0–100 from five criteria: topical fit (40), methodological fit (25), impact/venue (15), connectivity (10) and recency (10). Categories: 🔴 75+ must-read · 🟡 50–74 relevant · 🟢 30–49 radar. The score measures proximity to this newsletter's profile and prioritises reading — it is not a quality judgement on the individual paper.

In brief

About 25 scored papers, this week with **three 🔴 must-reads**, one of them title-based only. Focal points: **the duration of abrupt climate transitions**, **proxy system models for lake sediments**, and a **new tephra tie-point for the East African Rift**.

At the top: precisely dated speleothems from the Asian monsoon region show, according to a **Science study**, that Dansgaard-Oeschger warming transitions unfolded **over more than a century** — much more slowly than Greenland ice cores suggest. Also **PRYSM v2.0**, a proxy system model for lake sediments with a new runoff-isotope module (*GMD*), and an **offshore correlative of the Konso Silver Tuff** from the East African Rift (*QSR*; no abstract available yet, hence ⚠ title-based).

In the 🟡 block, among others: a **method comparison of vegetation reconstructions** for Last Glacial Europe (forest cover differs by up to 40% between methods), the **Taung child in its palaeoclimatic and biogeographic context**, **Lake Kinneret porewaters over 30 kyr** (⚠ title-based) and a **commentary on fragmented AMOC tipping** and what it means for early-warning signals.

Funding: **ERC Proof of Concept opens (12 Nov)**, **Horizon Europe palaeoclimate call opens (17 Nov)**, **ERC Consolidator (12 Jan 2027)**, **ICDP proposals (15 Jan 2027)**. Policy: the **WissZeitVG hearing on 4 Nov** showed irreconcilable positions; **final budget committee session on the 2027 federal budget on 12 Nov**, final vote on **27 Nov**.

Part A — Papers of the week

Covered are papers with a Crossref `created` date from 26 October 2026 onward that were not listed in a previous issue. For every entry with an accessible abstract (OpenAlex or Crossref), the assessment is based on that abstract; titles without an available abstract are marked ⚠ *title-based* and receive only minimum points for methodological fit, because the method is unknown.

Must-read (75+)

1. Glacial abrupt transitions in ocean circulation were much slower than ice-core records imply · *Science*, 2026 · **Score 87** Dansgaard-Oeschger events are thought to reflect rapid shifts of the Atlantic Meridional Overturning Circulation (AMOC); Greenland ice cores suggest warming transitions within a few decades or even just over ten years. Findings per abstract: **precisely dated speleothem δ¹⁸O records from the Asian monsoon region** show that the same warmings unfolded **over more than a century**. Isotope modelling suggests that Greenland records primarily capture a rapid atmospheric reorganisation triggered by initial sea-ice retreat in the northeastern North Atlantic, while the parallel AMOC recovery and hemispheric thermal adjustment proceeded much more slowly. Assessment: directly relevant for any analysis of transition durations, tipping points and early-warning signals in proxy series — "abrupt" depends on the archive and on which system component it records. *Topic 34 · Method 22 · Venue 15 · Connectivity 6 · Recency 10 (created 5 Nov 2026)* · DOI: 10.1126/science.aed4270

2. Calibration guidelines and a runoff-isotope module for lake proxy system modeling (PRYSM v2.0) · *Geoscientific Model Development*, 2026 · **Score 81** Proxy system models (PSMs) translate climate variables into proxy units and thus allow a mechanistic interpretation of lake-sediment data. PRYSM v2.0 adds a **simple runoff-isotope module** that can be adapted to any lake and, per abstract, reproduces the seasonal timing and magnitude at two mid-latitude lakes with contrasting morphometry, hydrology and mixing regimes. In model experiments the PSM is calibrated to a single observed profile and must predict lake-water temperature and δ²H — a typical data situation at barely monitored lakes; observations from the warm, ice-free season prove most informative. Assessment: closely related to lake water-balance and isotope models for closed basins; the calibration guidelines are transferable to unmonitored palaeolakes. *Topic 38 · Method 20 · Venue 9 · Connectivity 8 · Recency 6 (created 28 Oct 2026)* · DOI: 10.5194/gmd-19-10243-2026

3. A Konso Silver Tuff correlative found offshore the East African Rift · *Quaternary Science Reviews*, 2026 · **Score 79** · ⚠ *title-based* At the time of writing neither Crossref nor OpenAlex listed an abstract, and the publisher page could not be accessed. According to the title, a correlative of the **Konso Silver Tuff** (a tephra known from southern Ethiopia) was found **outside the East African Rift**, apparently in a marine archive. Assessment (with caveats): such a find could tie terrestrial rift sequences chronologically to marine records — potentially an important anchor for drill cores from southern Ethiopian rift basins. Please take age, archive and robustness of the correlation from the original paper. *Topic 40 · Method 8 · Venue 11 · Connectivity 10 · Recency 10 (created 9 Nov 2026)* · DOI: 10.1016/j.quascirev.2026.110364

Relevant (50–74)

4. Forest or tundra? How different vegetation reconstructions of Last Glacial landscapes in Europe may shape our perception of early human dispersal processes · *Journal of Quaternary Science*, 2026 · **Score 74** A comparison of four techniques — two pollen-based reconstructions and two vegetation models — for Europe between 70 and 20 ka. Findings

per abstract: **absolute forest cover differs by up to 40% between techniques**, enough to make the difference between boreal forest and tundra; the main reason is loosely defined forest-cover metrics. All techniques, however, reproduce ecosystem dynamics over time well on continental to regional scales; the authors recommend **relative forest-cover measures**. Methodologically transferable to any human–environment analysis built on reconstructed landscapes. *Topic 32 · Method 18 · Venue 8 · Connectivity 6 · Recency 10 (created 5 Nov 2026)* · DOI: 10.1002/jqs.70134

5. The Taung child and the biogeographic and palaeoclimatic drivers of southern African hominin diversity · *npj Palaeoecosystems*, 2026 · **Score 74** The authors reassess the site's chronological age, the fossil's dental and brain development and variation of the dentino-enamel junction, and conclude that *Australopithecus africanus* is a poorly defined taxon. Based on new palaeoclimatic and palaeoenvironmental evidence and South Africa's position at the Zambezian–southern African biogeographic boundary, they propose that **complex population dynamics (drift, hybridisation)** underlie hominin morphological diversity in the Plio-Pleistocene (a hypothesis, per abstract). Connected to questions of how climatic corridors and barriers structured hominin populations. *Topic 34 · Method 13 · Venue 11 · Connectivity 6 · Recency 10 (created 5 Nov 2026)* · DOI: 10.1038/s44513-026-00008-2

6. The evolution of Lake Kinneret during the past 30 kyr from porewaters in the lacustrine sediments · *Quaternary Science Reviews*, 2026 · **Score 73** · ⚠ *title-based* No abstract available. According to the title, the evolution of Lake Kinneret over 30 kyr is reconstructed from **porewaters in the lake sediments** — an approach that can address salinity and water balance of past lake stands directly. Thematically linked to the KIND drilling initiative (see Week 45). Findings with caveats. *Topic 36 · Method 8 · Venue 11 · Connectivity 8 · Recency 10 (created 5 Nov 2026)* · DOI: 10.1016/j.quascirev.2026.110357

7. Latitudinal Heterogeneity in South Asian Monsoon Changes Since the Last Glacial Maximum Revealed by Ganga–Brahmaputra Vegetation Record · *Paleoceanography and Paleoclimatology*, 2026 · **Score 72** A new marine pollen record from the Bay of Bengal, compared with leaf-wax $\delta^{13}C$/δD and foraminiferal $\delta^{18}O$ from the same core. Findings per abstract: an open, drier and colder landscape in MIS 2, tropical forests in MIS 1; Holocene large-scale drying with more pronounced winter dryness after 8 ka. Comparison with records farther south reveals a **persistent north–south hydroclimate gradient** — according to the authors pointing to additional modulation by orography and moisture recycling beyond ITCZ control. *Topic 30 · Method 15 · Venue 11 · Connectivity 6 · Recency 10 (created 3 Nov 2026)* · DOI: 10.1029/2026pa005425

8. Late Quaternary drainage reorganization of the Sutlej River system at the Himalaya mountain-foreland transition · *Communications Earth & Environment*, 2026 · **Score 72** Geomorphic mapping, subsurface stratigraphy and OSL dating where the Sutlej exits the Himalayan front. Findings per abstract: a previously misidentified uplifted surface, abandoned palaeovalleys and terraces; drainage reorganisation via bifurcation and avulsion occurred **between > 60 and 15 ka**, tectonically controlled — long before the mid-Holocene decline of the Harappan Civilisation. An example of how fluvial archives help separate tectonic from climatic control. *Topic 30 · Method 15 · Venue 11 · Connectivity 6 · Recency 10 (created 7 Nov 2026)* · DOI: 10.1038/s43247-026-04204-5

9. Fragmentation of AMOC tipping · *Dialogues on Climate Change*, 2026 · **Score 69** Commentary (no new data): ocean models indicate that the AMOC may undergo several partial collapses before a full collapse; in a conceptual box model this **fragmentation complicates the**

interpretation of early-warning signals. The piece discusses whether these intermediate transitions survive in more detailed models — particularly because the convection parameterisation can introduce additional, possibly spurious branches in the bifurcation diagram — and calls for convergence tests. Conceptually relevant for tipping-point analyses of palaeorecords. *Topic 30 · Method 18 · Venue 5 · Connectivity 6 · Recency 10 (created 4 Nov 2026)* · DOI: 10.1177/29768659261502897

10. Sequential high-density radiocarbon dating to detect invisible event layers in marine sediments · *Radiocarbon*, 2026 · **Score 67** Continuous ^{14}C dating of total organic carbon at 0.5–1 cm intervals: age reversals mark, per abstract, event layers from submarine slope failures rather than flood input; after a foraminiferal age model, the layers correlate with historical natural disasters since the 15th century. Methodologically interesting for detecting visually inconspicuous event layers, including in palaeoflood archives. *Topic 25 · Method 18 · Venue 8 · Connectivity 6 · Recency 10 (created 9 Nov 2026)* · DOI: 10.1017/rdc.2026.10255

Radar

- **Can multi-glacier moraine analysis constrain paleoclimate reconstructions? Evidence from 3D glacier modeling in the Cordillera Real, Bolivia** · *Climate of the Past* · **67** · Full-Stokes model of neighbouring glaciers: geometry and hypsometry cannot separate temperature from precipitation; ice thickness could, but only with very accurate palaeo-volumes. Open access. DOI: 10.5194/cp-22-1941-2026 · created 5 Nov 2026
- **Recent Slowing Arctic-Amplified Warming Drives an Unprecedented Water Recovery in the World's Most Voluminous Freshwater Lake** · *Geophysical Research Letters* · **67** · 343-year tree-ring reconstruction of Lake Baikal in- and outflow: ~90% decline in the 20th century, ~80% increase over the last decade. DOI: 10.1029/2026gl124400 · created 3 Nov 2026
- **Long-term Lacustrine Evolution on the Tibetan Plateau Inferred from Ostracods in ICDP NamCore Sediments** · *Acta Micropalaeontologica Sinica* · **67** · ⚠ *title-based* · Ostracods from the ICDP Nam Co core. DOI: 10.3724/micropals-2026111 · created 9 Nov 2026
- **Water Vapor Buoyancy and the African Easterly Jet** · *Geophysical Research Letters* · **65** · The meridional moisture gradient diagnostically weakens the African Easterly Jet by ~30%; the effect strengthens with warming in CMIP6 but is neglected by some models. DOI: 10.1029/2026gl124156 · created 6 Nov 2026
- **Post-LGM fluvial and hydroclimatic variability in the Haryana plains** · *Quaternary Science Reviews* · **65** · ⚠ *title-based*. DOI: 10.1016/j.quascirev.2026.110361 · created 3 Nov 2026
- **Enhanced pedogenic weathering of aeolian dust as a Pleistocene carbon sink on the Chinese Loess Plateau** · *Geology* · **64** · A new method separates pedogenic weathering from sorting effects (6-Myr sequence): the weathering-driven carbon sink was higher in the dry, cool Pleistocene than in the Pliocene. DOI: 10.1130/g54703.1 · created 5 Nov 2026
- **Middle-to-late Holocene contraction of Gala Co on the southern Tibetan Plateau driven by a weak…** · *Palaeogeography, Palaeoclimatology, Palaeoecology* · **62** · ⚠ *title-based* · Holocene lake-level decline. DOI: 10.1016/j.palaeo.2026.114328 · created 4 Nov 2026
- **Rate-induced tipping of ice sheets due to visco-elastic Earth response under idealized conditions** · *The Cryosphere* · **61** · Idealised ensembles: the interaction of ice dynamics and visco-elastic Earth response affects the tipping dynamics of an Antarctic-type ice-sheet–shelf system. DOI: 10.5194/tc-20-6131-2026 · created 28 Oct 2026

- **High- and low-latitude forcing on precipitation variations over the central Qilian Mountains since 18.5 ka** · *Quaternary Science Reviews* · **60** · Multi-proxy loess record (carbonate $\delta^{18}O$ and $\delta^{13}C$) on the interplay of the East Asian summer monsoon and the westerlies. DOI: 10.1016/j.quascirev.2026.110343 · created 26 Oct 2026
- **Energetics of hominin group locomotion: Modeling coordinated mobility using the Engare Sero footprint assemblage** · *Journal of Human Evolution* · **58** · Cost-of-transport modelling of Late Pleistocene footprints (Tanzania): a group mean of 1.41 m/s lies near the shared energetic optimum. DOI: 10.1016/j.jhevol.2026.103912 · created 7 Nov 2026
- **Recognizing post-entrapment alteration for robust speleothem fluid inclusion microthermometry** · *EGUsphere* (preprint) · **57** · Cryptically altered inclusions bias homogenisation temperatures upward; the first quartile is proposed as a robust estimator. DOI: 10.5194/egusphere-2026-6315 · created 4 Nov 2026
- **Ahead of its time: an interdisciplinary investigation of Lake Zurich's sediment fill (Zübo 80 project)** · *E&G Quaternary Science Journal* · **56** · A look back at an early interdisciplinary lake-drilling project of the 1980s. DOI: 10.5194/egqsj-75-191-2026 · created 5 Nov 2026
- **Silicon redistribution and promoted low-latitude diatom growth after Atlantic circulation recovered from deglacial meltwater discharge** · *EGUsphere* (preprint) · **55** · AWI-ESM2.1 experiments on the silicon cycle at the LGM and during deglaciation. DOI: 10.5194/egusphere-2026-6527 · created 4 Nov 2026
- **The first filling of the Grand Ethiopian Renaissance Dam (GERD): hydrological and operational insights** · *Communications Earth & Environment* · **52** · Model and satellite reconstruction of the first filling 2020–2024; thanks to very wet years few downstream impacts, while floodplain agriculture in Sudan declined by 4–50%. DOI: 10.1038/s43247-026-04201-8 · created 7 Nov 2026
- **Seismic monitoring of a diamond core drill bit during ICDP expedition 5071** · *Journal of Applied Geophysics* · **51** · Seismic-while-drilling in the ICDP project DIVE: no drill-bit signal detected; a feasibility study. DOI: 10.1016/j.jappgeo.2026.106614 · created 6 Nov 2026

Not listed or only noted: The *Nature* news item on 26,000-year-old cave art and the *Science* commentary "Start cryosphere flood warnings at the source" are not primary articles. Not included: numerous off-topic hits from topic queries ("early warning", "Ethiopia", "fluvial" in medical, economic or engineering contexts) and pre-Quaternary papers without profile relevance.

Part B — Funding & calls

ERC Proof of Concept 2027 — opens 12 Nov 2026, first cut-off 10 Mar 2027 (per a summary of Work Programme 2027; check the ERC site before planning). Source: Hezelburcht — ERC Work Programme 2027.

Horizon Europe — "Palaeoclimate science for a better understanding of Earth system dynamics" (HORIZON-CL5-2027-01-D1-08). Opens **17 Nov 2026**, deadline **4 Mar 2027**, Research and Innovation Action, about €14 million for roughly two projects (as of previous issue). Source: Topic page.

ERC Consolidator Grant 2027 — deadline 12 Jan 2027. Usually 7–12 years after the PhD. Source: ERC — Consolidator Grant.

ICDP — 2027 proposal cycle. Recurring **15 January** deadline for pre-, workshop, full and fast-track proposals; the 2027 date is not formally confirmed. Source: ICDP — Proposal Preparation.

DFG — Open-access publication funding. Institutional proposals first due **4 May 2027** (as of previous issue). Source: DFG — Information for Researchers No. 68.

No new profile-relevant calls with a deadline were found this week; the entries above are carried over. The ICDP AGU travel grants (deadline 8 Nov) have closed.

Part C — Science policy (DE/EU)

WissZeitVG — research committee hearing on 4 Nov 2026: positions remain irreconcilable. According to *Forschung & Lehre*, Birgitta Wolff (rector, Univ. Wuppertal) praised the minimum standards for first contracts and warned against a return to general fixed-term rules; Patrick Cramer (Max Planck Society) stressed the balance between reliable career paths and flexibility. Amrei Bahr (Univ. Stuttgart) said the coalition agreement's promise was not kept and called, among other things, for lifting the collective-bargaining block, a maximum quota of fixed-term contracts and removing postdocs from the law. Sonja Bolenius (DGB) and Andreas Keller (GEW) welcomed extending parental and care-leave provisions to grant-funded contracts but criticised the missing sharper definition of "qualification" and ultimately argued for abolishing the special fixed-term regime. GEW cites 84% of academic staff at universities on fixed-term contracts. No date for the 2nd/3rd reading was publicly available at the time of writing. Assessment: for grant-funded postdoc positions, minimum contract durations and the family provisions remain the tangible changes; whether the coalition tightens the bill after the hearing is open. Sources: Forschung & Lehre (4 Nov 2026) · GEW (4 Nov 2026) · Table.Briefings (5 Nov 2026, paywall).

German federal budget 2027 — final budget committee session on 12 Nov, final vote on 27 Nov. Changes to individual budgets, including BMFTR project funding and DFG allocations, are only possible in the final session; government draft: BMFTR core budget about €22 billion, nominally about +0.7% (as of previous issue). Sources: Bundestag — 2027 budget timeline · Forschung & Lehre — Only a small increase for the BMFTR core budget.

EU budget 2027 — Council/Parliament agreement deadline around 16 Nov. The Council had proposed €12.6 billion for Horizon Europe (−€231.5 million versus the Commission), while Parliament voted for an increase (see Week 45). A negotiation outcome could not be verified at the time of writing. Separately, the Commission reports an **additional €500 million for Horizon Europe in 2026 and 2027** (date and details of the release could not be fully checked). Sources: Science|Business — Council proposes €231m cut · European Commission — IP/26/2004.

Part D — Community & dates

- **ERC Proof of Concept** — opens **12 Nov 2026**.
- **German federal budget 2027** — final committee session **12 Nov**, final vote **27 Nov 2026**.
- **EU budget 2027** — Council/Parliament agreement deadline around **16 Nov 2026**.
- **Horizon CL5 palaeoclimate call** — opens **17 Nov 2026**, deadline **4 Mar 2027**.
- **AGU Fall Meeting 2026** — **7–11 Dec 2026**, San Francisco. Source: AGU.

- **ERC Consolidator Grant 2027** — deadline **12 Jan 2027**.
- **ICDP** — proposal deadline expected **15 Jan 2027**.
- **EGU General Assembly 2027** — **4–9 Apr 2027**, Vienna; abstract submission open (deadline usually mid-January, not verified). Source: PAGES Newsletter October 2026.
- **PAGES dates 2027** — PALSEA-Next **24–26 Jun 2027** (Dartmouth College); **PMIP conference 30 Aug – 3 Sep 2027** (Rio de Janeiro). Source: PAGES Newsletter October 2026.

Methods & sources

Paper screening follows the fixed criteria (topical fit, methodological fit, impact/venue, connectivity, recency) and excludes DOIs already listed in a previous issue. Funding, policy and dates are based on the sources linked in Parts B to D.

How this issue was produced. Direct network access from the computing environment was blocked. The **Crossref REST API** (ISSN queries for about 30 journals, including QSR, Climate of the Past, Quat. Sci. Adv., GPC, EPSL, Boreas, JQS, Quat. Res., P³, Nature, Science, PNAS, Nature Commun., Nature Geosci., Sci. Adv., Commun. Earth Environ., GRL, G-cubed, Paleoceanogr. Paleoclimatol., Quat. Int., The Holocene, Geology, J. African Earth Sci., J. Human Evol., Radiocarbon; Copernicus prefix including EGUsphere; 15 topic queries; filter `from-created-date: 2026-10-26`) and **OpenAlex** (abstracts by DOI from the `abstract_inverted_index`) were queried through the integrated browser interface (`fetch()`). For several very recently registered Elsevier titles no abstract was available yet; ScienceDirect could not be accessed.

What this means for coverage. Where neither OpenAlex nor Crossref provides an abstract, the entry is marked ⚠ *title-based*; no findings beyond the title were invented, and methodological fit receives minimum points. Dates are Crossref `created` dates (registration), which may differ from the online publication date by a few days. Topic queries return a lot of off-topic noise; pre-selection used keywords followed by manual review, so individual relevant papers outside the core journals may be missing. Listed papers were checked for co-authorship by the publisher of this digest; there is none this week.

GeoNews is an automatically generated digest published by Dr. Markus L. Fischer, Palaeoclimate Dynamics group, University of Potsdam. The thematic profile follows his research interests; selection is made by a rule-based workflow, not an editorial team. The digest is an independent literature overview with no affiliation to the publishers or funders mentioned and does not constitute a recommendation or advice. Entries may contain errors; when in doubt, the linked original source prevails. Corrections and comments are welcome.